2026-2027 学年第一学期

《高级语言程序设计(C++语言)》

实验报告

实验名称：实验三　函数	
实验日期：2026 年 11 月 1 日	实验地点：
学　　号：	姓　　名：
学　　院：	专　　业：
指导老师：	

一、实验目的

掌握函数的定义、调用、递归与重载。

二、实验环境

Windows 11、VS Code、GCC 16.2.0。

三、实验过程及结果

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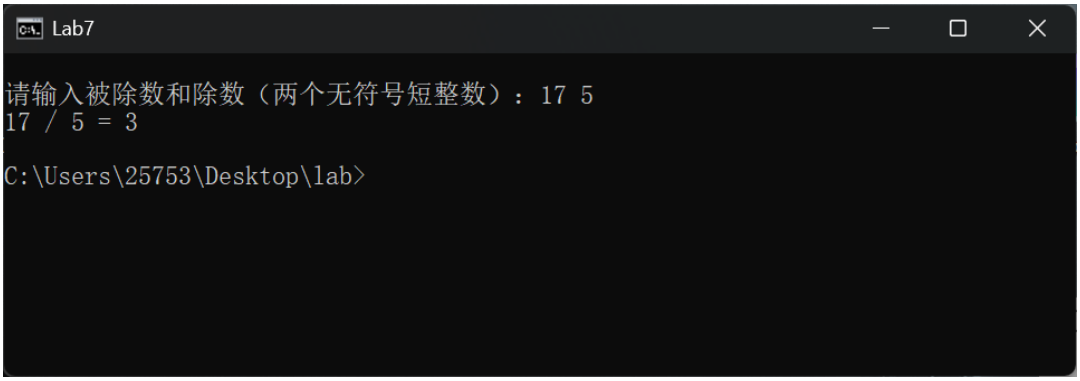
```
#include <iostream>
using namespace std;

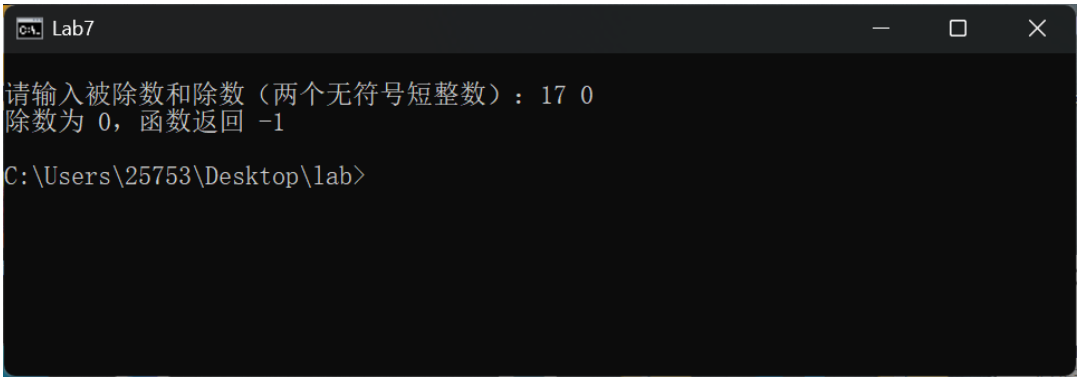
short int divide(unsigned short int a, unsigned short int b)
{
    if (b == 0)
        return -1;
    return static_cast<short int>(a / b);
}

int main()
{
```

```
    unsigned short int a, b;
    cout << "请输入被除数和除数（两个无符号短整数）: ";
    cin >> a >> b;

    short int r = divide(a, b);
    if (b == 0)
        cout << "除数为 0，函数返回 " << r << endl;
    else
        cout << a << " / " << b << " = " << r << endl;
    return 0;
}
```





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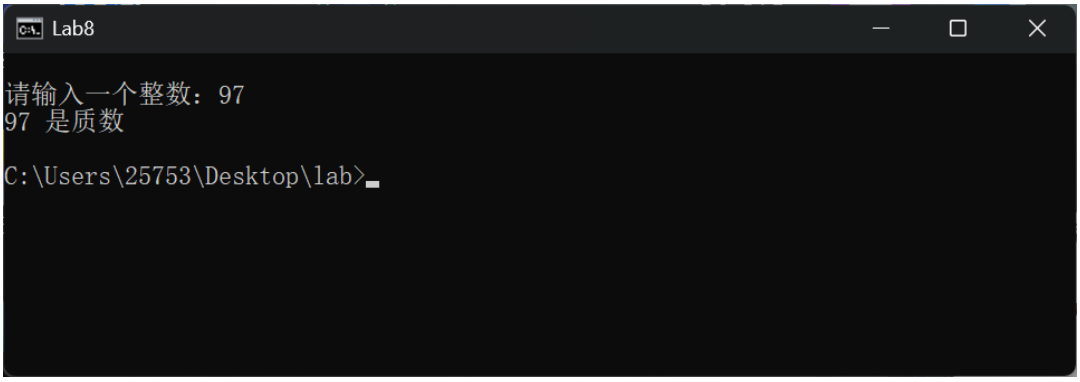
```
#include <iostream>
using namespace std;

bool isPrime(int n)
{
    if (n < 2)
        return false;
    for (int i = 2; i <= n / i; i++)
        if (n % i == 0)
            return false;
    return true;
}

int main()
{
    int n;
    cout << "请输入一个整数: ";
    cin >> n;

    if (isPrime(n))
        cout << n << " 是质数" << endl;
    else
```

```
        cout << n << " 不是质数" << endl;
    return 0;
}
```



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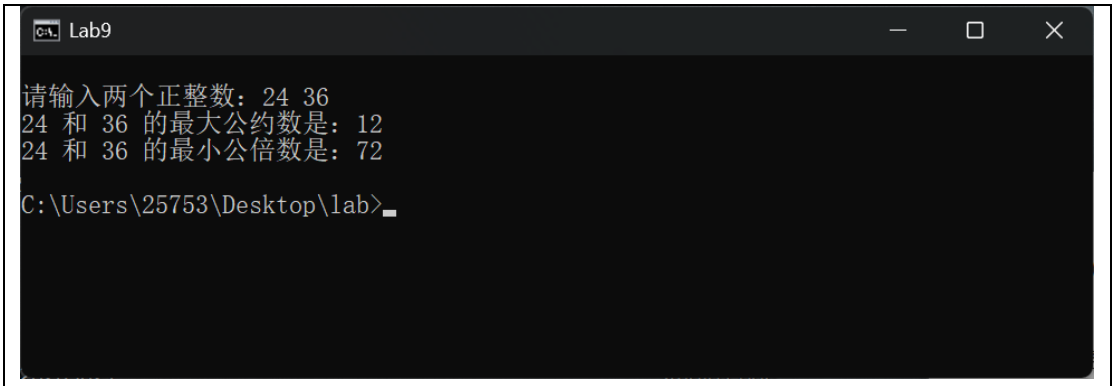
```
#include <iostream>
using namespace std;

int gcd(int a, int b)
{
    while (b != 0)
    {
        int t = a % b;
        a = b;
        b = t;
    }
    return a;
}

int lcm(int a, int b)
{
    return a / gcd(a, b) * b;
}

int main()
{
    int a, b;
    cout << "请输入两个正整数：";
    cin >> a >> b;

    cout << a << " 和 " << b << " 的最大公约数是：" << gcd(a, b) << endl;
    cout << a << " 和 " << b << " 的最小公倍数是：" << lcm(a, b) << endl;
    return 0;
}
```



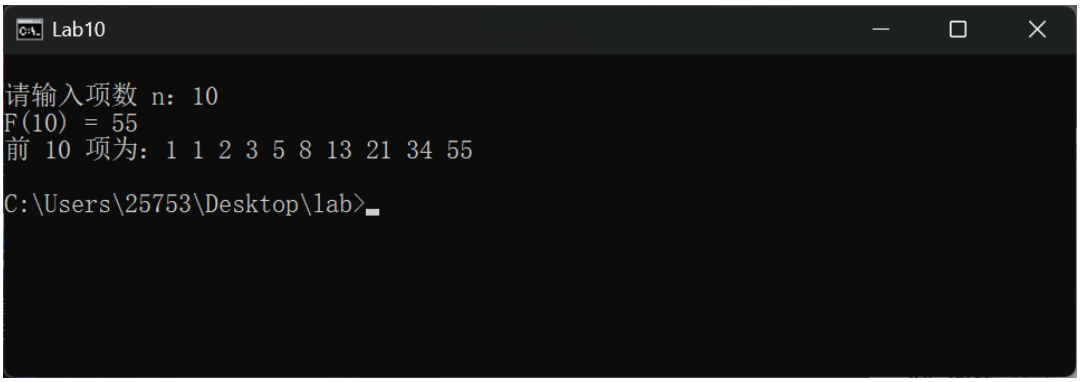
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```
#include <iostream>
using namespace std;

long long fib(int n)
{
    if (n <= 2)
        return 1;
    return fib(n - 1) + fib(n - 2);
}

int main()
{
    int n;
    cout << "请输入项数 n：";
    cin >> n;

    cout << "F(" << n << ") = " << fib(n) << endl;
    cout << "前 " << n << " 项为：";
    for (int i = 1; i <= n; i++)
        cout << fib(i) << " ";
    cout << endl;
    return 0;
}
```



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```
#include <iostream>
using namespace std;

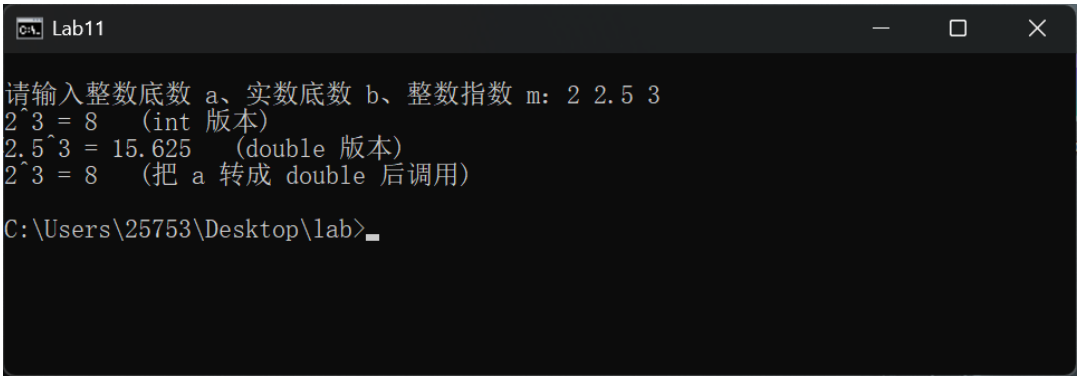
int getPower(int x, int y)
{
    if (y < 0)
        return 0;
```

```
    if (y == 0)
        return 1;
    return x * getPower(x, y - 1);
}

double getPower(double x, int y)
{
    if (y == 0)
        return 1.0;
    if (y > 0)
        return x * getPower(x, y - 1);
    return 1.0 / getPower(x, -y);
}

int main()
{
    int a, m;
    double b;
    cout << "请输入整数底数 a、实数底数 b、整数指数 m: ";
    cin >> a >> b >> m;

    double r = getPower(static_cast<double>(a), m);
    cout << a << "^" << m << " = " << getPower(a, m) << "   (int 版本)" << endl;
    cout << b << "^" << m << " = " << getPower(b, m) << "   (double 版本)" << endl;
    cout << a << "^" << m << " = " << r << "   (把 a 转成 double 后调用)" << endl;
    return 0;
}
```



四、实验总结（本次实验的收获、体会、经验和教训）

接口比函数体重要，边界数据须实测。